Standard for the Description of Rice Germplasm Resources

1 Scope

This specification stipulates the descriptors for rice germplasm resources and their grading criteria.

This Code applies to the collection, organization, and preservation of rice germplasm resources; the development of data standards and data quality control specifications; and the establishment of database and information-sharing network systems.

2 Normative References

The provisions in the following documents shall form part of this specification by reference therein. For dated reference documents, all subsequent corrigenda (excluding errata) or revised editions shall not apply to this specification. However, parties to agreements concluded in accordance with this specification are encouraged to consider whether the latest editions of these documents may be used. For undated reference documents, the latest edition shall apply to this specification.

ISO3166 Codes for the Representation of Names of Countries

GB/T 2659: Country and Region Name Codes

GB/T 2260: Administrative Division Codes of the People's Republic of China

GB/T 12404 Code for Unit Affiliation

GB/T 3543.6-1995: Test Procedure for Moisture Determination in Crop Seeds

GB/T 5495-1985: Test Method for the Hulling Rate of Grains and Oil Crops (Rice)

GB/T 5492-1985: Inspection of Grains and Oil Crops—Methods for the Identification of Color, Odor, and Taste

GB/T 17891-1999 High-quality rice

GB/T 15683-1995 Determination of amylose content in rice

GB/T 2905-1982 Method for Determination of Crude Protein in Seeds of Cereal and Leguminous Crops

GB/T 5519-1988: Determination of the thousand-kernel weight of grains and oilseeds

GB/T 4801-1984 Determination of Lysine in Cereal Grains

GB/T 2906-1982 Method for Determination of Crude Fat in Seeds of Cereal and Oilseed Crops

NY/T 11-1985 Determination of Crude Starch in Grain Crops

NY/T 83-1988: Method for Determination of Quality

NY/T-593-2002 Quality of Edible Rice Varieties

NY/T 594-2002 Edible Japonica Rice

NY/T 595-2002 Edible Indica Rice

GB/T 195557.7-2004 Guidelines for the Assessment of Specificity, Consistency, and Stability of New Plant Varieties: Rice—Standard Evaluation System for Rice (SES), IRRI, 2002

GB/T 15790-1995 Specifications for the Monitoring and Investigation of Rice Blast

GB/T 15791-1995 Specifications for the Monitoring and Investigation of Wood Rot

GB/T 15792-1995 Specifications for Monitoring and Surveying Rice Two-Looper Stem Borer

GB/T 15794-1995 Specifications for the Monitoring and Survey of Rice Leafhoppers

3 Terms and Definitions

3.1 Paddy

Gramineae (Graminear*),* the genus Oryza. The genus Oryza comprises two cultivated species, namely Asian rice *(Oryza sativa* L.) and African rice (*Oryza glaberrima Steud.),* as well as 20–25 wild species. Rice (O *ryza sativa* L.) has a chromosome number of 2n= 24.

3.2 Rice Germplasm Resources

Wild rice resources, local varieties, selected and bred varieties, lines, hybrid rice resources, genetic materials, mutants, and others.

3.3 Basic Information

Basic descriptive information on rice germplasm resources, including the nationally unified identification number, germplasm name, scientific name, place of origin, germplasm type, and other relevant details.

3.4 Morphological Characteristics and Biological Properties

Phenological stages, botanical morphology, yield traits, and other characteristic traits of rice germplasm resources.

3.5 Quality Characteristic

The milling quality, appearance quality, cooking and edible quality, and nutritional quality of rice germplasm resources, among other characteristics. These include the brown rice yield rate, polished rice yield rate, total polished rice yield rate, brown rice shape, chalky-white grain rate, chalky-white particle size, chalkiness, transparency, gelatinization temperature, gel consistency, amylose content, protein content, lysine content, fat content, aroma, and other parameters.

3.6 Stress resistance

The adaptability or resistance of rice germplasm resources to various abiotic stresses, including drought tolerance, salt tolerance, alkaline tolerance, cold tolerance, and heat tolerance.

3.7 Resistance to pests and diseases

The adaptability or resistance of rice germplasm resources to various biological stresses, including resistance to rice blast, white leaf blight, damping-off, bacterial stripe disease, white-backed planthopper, brown planthopper, rice thrips, two-leaf stem borer, and three-leaf stem borer, among others.

4 essential information

4.1 Nationwide Unified Number

The unique identifier for germplasm resources. A unified national number is assigned based on the variety type and place of origin. For example, developed varieties begin with ZD; local varieties from Beijing begin with 01; introduced varieties begin with WD; hybrid rice combinations begin with H; sterile lines begin with A; maintainer lines begin with B; and restorer lines begin with R. 4.2 Germplasm Bank Numbering

The accession number of the rice germplasm in the National Long-term Crop Germplasm Resource Bank.

4.3 Introduction No.

The designation assigned to rice germplasm when it was introduced from abroad.

4.4 Collection ID

The identification number assigned to rice germplasm during its field collection.

4.5 Germplasm Designation

The Chinese name for rice germplasm.

4.6 Genetic Resource Foreign Name

The foreign-language name of the introduced germplasm or the Chinese Pinyin name of the domestic germplasm.

4.7 Field Name

Gramineae (Grasses*).*

4.8 generic name

Oryza (the genus Oryza).

4.9 scientific name

Rice (Asian cultivated rice) has the scientific name Oryza *sativa L.*

4.10 country of origin

The name of the country of origin, the name of the region, or the name of the international organization of the rice germplasm.

4.11 Country of Origin

The native province of the rice germplasm.

4.12 Country of Origin

The names of the counties, townships, and villages of origin of rice germplasm.

4.13 Height

Altitude of the native origin of rice germplasm. Unit: m.

4.14 Longitude

The longitude of the origin of rice germplasm is expressed in degrees and minutes, using the format DDDFF, where DDD represents degrees and FF represents minutes.

4.15 Latitude

The latitude of the origin of rice germplasm, expressed in degrees and minutes, follows the format DDFF, where DD represents degrees and FF represents minutes.

4.16 Origin

For rice germplasm introduced from abroad, provide the name of the country, region, or international organization from which the germplasm originated; for domestic germplasm, provide the name of the province or county from which it originated.

4.17 Preserving Entity

The name of the original institution where the rice germplasm was preserved prior to its submission to the National Long-term Crop Germplasm Resource Bank.

4.18 Save Unit Number

The germplasm accession number assigned by the original germplasm preservation institution for the rice germplasm.

4.19 genealogy

The genetic relationships among rice breeding varieties (lines).

4.20 Breeding Unit

The name of the institution or individual responsible for the breeding and development of rice varieties (lines).

4.21 Age at maturation

The year in which a rice variety (line) was successfully developed.

4.22 Breeding Method

Breeding methods for rice varieties (lines), including systematic breeding, hybrid breeding, mutagenic breeding, and ploidy breeding.

Varietal breeding, molecular breeding, and others.

4. 23 germplasm types

Rice germplasm types are divided into eight categories.

1 Wild resources

2 local variety

3 Selected breeding varieties

4 Strain

5 Hybrid rice resources

6 genetic stocks

7 mutant

8 other

4.24 picture

Image filename for rice germplasm. The image format is.jpg.

4.25 observation place

Name of the observation site for the morphological characteristics and biological traits of rice germplasm.

5 Morphological Characteristics and Biological Properties

5.1 Subspecies Type

It comprises two subspecies: indica rice and japonica rice.

1 nonglutinous rice

2 round-grained nonglutinous rice

5.2 Hydric and xeric tolerance

It is divided into paddy rice and upland (dryland) rice.

1 paddy

2 Upland rice

5.3 Adhesiveness

The viscosity and stickiness of the endosperm.

1 Glutinous rice

2 glutinous rice

5.4 Photothermal property

Differences in sensitivity to light and temperature.

1 early season rice

2 semilate rice

3 late rice

5.5 Mature-phase

The overall performance of rice climate-ecological types in terms of ripening characteristics.

1 precocity

2 Medium-mature

3 late-maturing

5.6 sowing time

The seed sowing date used when assessing the morphological characteristics and biological traits of rice germplasm. It is expressed as "year month day," in the format "YYYYMMDD."

5.7 Booting Stage

The date at which 10% of the rice ears have emerged during the heading stage across the entire district. The heading standard for each ear is defined as the point at which the ear tip protrudes 3 cm beyond the leaf sheath. The date shall be expressed as "year month day" in the format "YYYYMMDD."

5.8 Heading Period

50% of the rice ears in the entire region reach heading date. This should be expressed as "year month day," in the format "YYYYMMDD."

5.9 Grain Maturity Period

80% of the rice ears in the entire region reach heading date. The date is expressed in "year month day" format, "YYYYMMDD."

5.10 maturation period

More than 90% of the entire region records the date when the kernels have reached full maturity. The date is expressed as "year month day," in the format "YYYYMMDD."

5.11 Whole reproductive period

The number of days required from the budding date to the maturity date, denoted as d.

5.12 Plant Height

The distance from the ground to the top of the tallest panicle in a mature rice plant, excluding the awn; expressed in cm.

5.13 Long Stem

The length of the rice plant from the ground level to the panicle collar, expressed in cm.

5.14 穗 Length

After rice has matured, the distance from the culm joint to the tip of the panicle (excluding the awn), expressed in cm.

5.15 Number of Grains per Spike

The total number of spikelets per spike equals the sum of the number of filled grains, empty or shriveled grains, and shed grains. The unit is grains.

5.16 Number of effective panicles

The total number of effective panicles per plant. Any panicle that has emerged and contains five or more grains is considered an effective panicle.

5.17 Sui extraction degree

The distance from the leaf cushion to the spike collar (Figure 1).

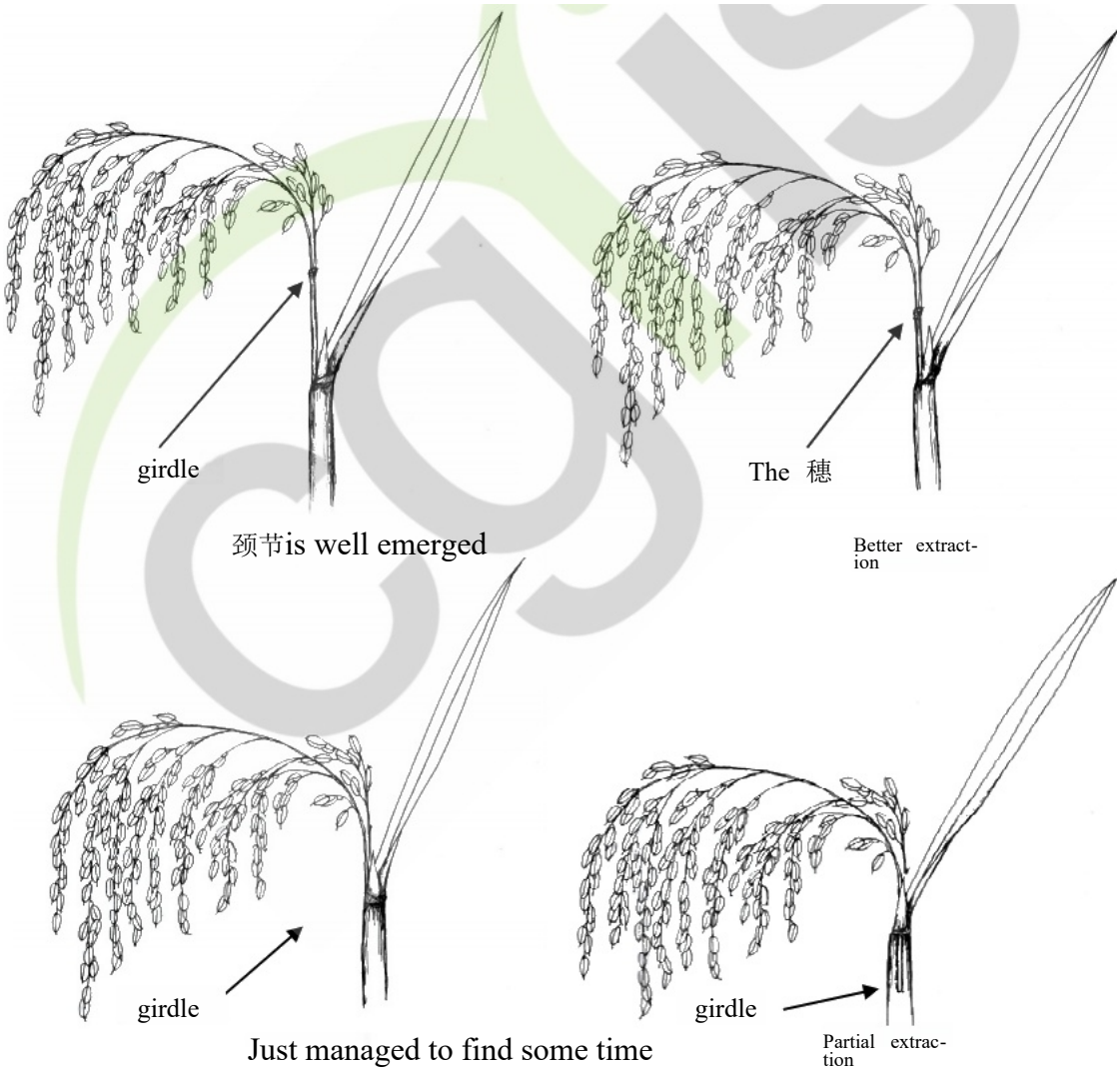




Figure 1:穗抽出度

girdle ————

Tight wrap

1 Good extraction

3 Better extraction

5 Just managed to find some time

7 Partial extraction

9 Tight wrap

5.18 穗型

The dispersion and density of rice panicle stalks (Figure 2).

1 concentrated

2 intermedius type

3 diffuse

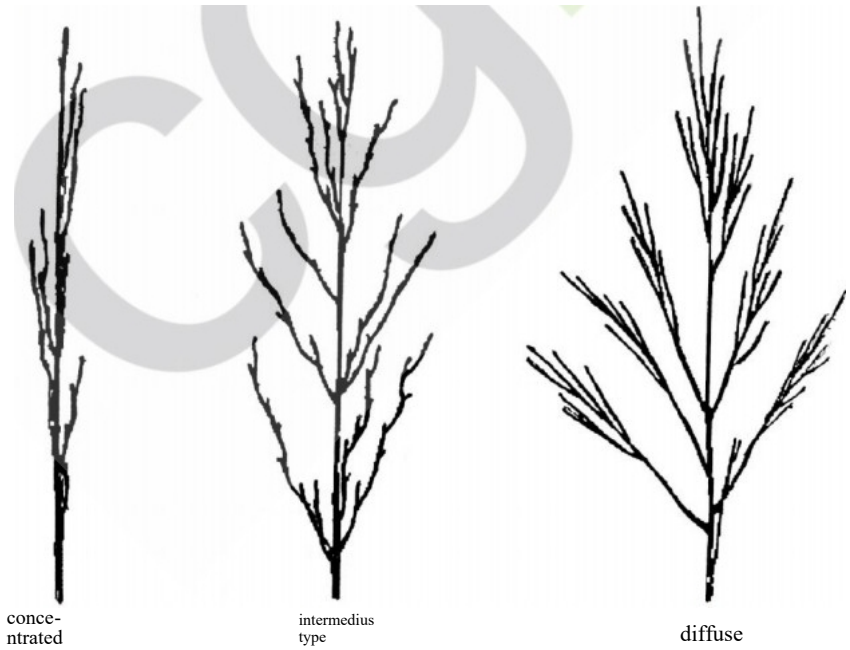


Figure 2: Spike Type

5.19 Second-Order Branch

The occurrence of spikelets attached to secondary branch peduncles on rice ears (Figure 3).

1 not have
5
Less 7
More
9 gather



Figure 3: Situation of Secondary Branch-Patent Spikelets

5.20 Sui-Li Shape

The erectness of the panicle axis. This is assessed based on the angle formed between the line connecting the leaf sheath to the panicle tip and the extension line of the stem (Figure 4).

1 up-right

5 semi-erect

7 winding

9 hang down



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Graph 4 Sui-li shape

5.21 Stability rate

The proportion of the average number of kernels relative to the average total number of florets per spike, expressed as a percentage.

5.22 thousand-seed weight

The weight of 1,000 plump rice grains (determined by water-washing and sedimentation), expressed in g.

5.23 Grain length

The length from the base of the glume sheath at the lowermost part of the grain to the tip of the longest inner glume or outer glume (glume apex), expressed in mm.

5.24 Grain width

The distance between the widest points on the inner and outer sides of the beak. Unit: mm.

5.25 Grain thickness

The distance between the inner and outer surfaces at the thickest point in between. Unit: mm.

5.26 Grain shape

Evaluate based on the ratio of grain length to width (Figure 5).

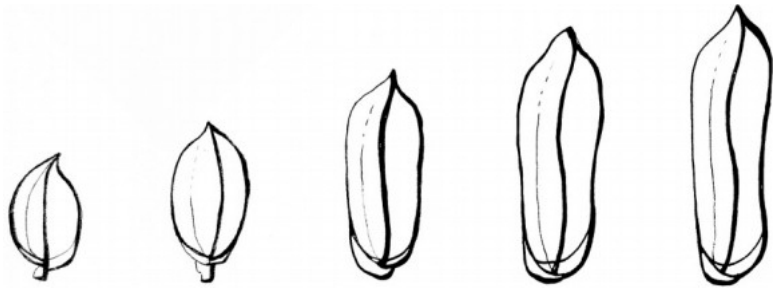
1 Short oval

3 broad-ovate

5 oval

7 Medium to long

9 elongate



Short and rounded, broad oval circular oval-shaped microsilar China Fine shape

Figure 5: Grain Shape

5.27 Brown rice length

Length of brown rice (dehulled grains). Unit: mm.

5.28 Brown rice width

The maximum width at the widest point of the brown rice. Unit: mm.

5.29 Brown rice thickness

The maximum thickness of the coarse rice layer, expressed in mm.

5.30 Brown rice shape

Evaluation based on the shape of the brown rice (Figure 6).

1 Nearly circular

3 oval

5 semi-spindle-shaped

7 Spindle-shaped

9 sharp spindle-shaped

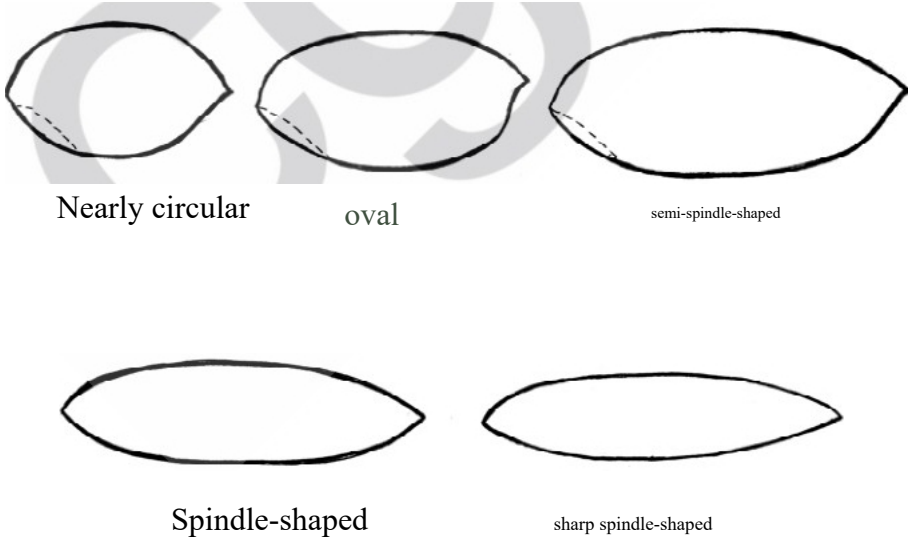


Figure 6: Shape of Brown Rice

5.31 Seed Coat Color

The color of the bran layer of brown rice.

1 white
2 red
3 brown
4 purple
5 black

5.32 Bud Sheath Color

The color of the shoot bud sheath.

1 Colorless
2 lilac colour
3 Deep purple

5.33 Leaf sheath color

The color of the leaf sheath.

1 yellow
2 green
3 purple

5.34 Leaf color

The color of the leaves.

1 buff
2 yellow spotting
3 Green and white
4 Light green
5 green
6 Deep green
7 Peripheral purple
8 Purple spots
9 purple

5.35 Leaf pubescence

The distribution of trichomes on the leaf surface.

1 not have

5 Dredging

7 China

9 secret

5.36 Blade curling degree

The degree of leaf curling.

1 No curl, or very slight curl

2 Main Volume

3 Anti-roll

4 heliciform

5.37 Sword leaf length

The distance from the petiole base to the leaf tip, expressed in cm.

5.38 Blade width

The distance at the widest point of the sword leaf blade. Unit: cm.

5. 39 Blade Angle Angle

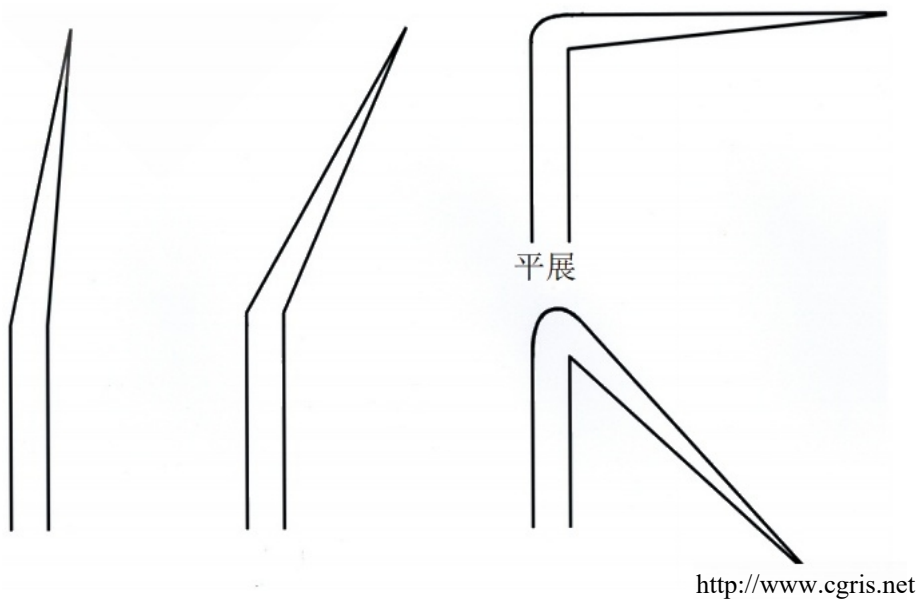
The angle formed between the blade of the sword leaf and the extension line of the stem (Figure 7).

1 up-right

5 intermedius type

7 Flat

9 披垂



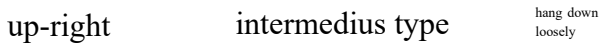


Figure 7: Blade Angle

5.40 Length of the second leaf after folding

Distance from the petiole junction to the leaf tip of the penultimate leaf. Unit: cm.

5.41 Width of the Inverted Second Leaf

Distance between the widest points of the penultimate leaf. Unit: cm.

5.42 Reverse Second Leaf Angle

The angle formed between the line connecting the tip of the penultimate leaf and the leaf collar, and the main stem.

1 up-right

5 Flat

9 hang down

5.43 Color of the auricles

The color of the leaf margins.

1 Colorless

2 yellow

3 green

4 purple

5.44 Ligule color

The color of the ligule.

1 No leaf tip

2 white

3 Purple line

4 purple

5.45 Ligule shape

The shape of the apex of the ligule (Fig. 8).

1 No leaf tip

2 pointed to gradually pointed

3 Bifurcation

4 truncation



pointed to gradually pointed Bifurcation truncation

Figure 8: Tongue Shape

5.46 Pedicell color

The surface color of the cotile.

1 green

2 purple

5.47 Leaf node **color**: the color of the leaf node.

1 No (white) color

2 green

3 purple

5.48 Number of leaves on the main stem

The total number of leaves produced over the entire lifespan of the main stem.

5.49 Stem angle

The angle between the stem and the vertical line relative to the ground (Figure 9).

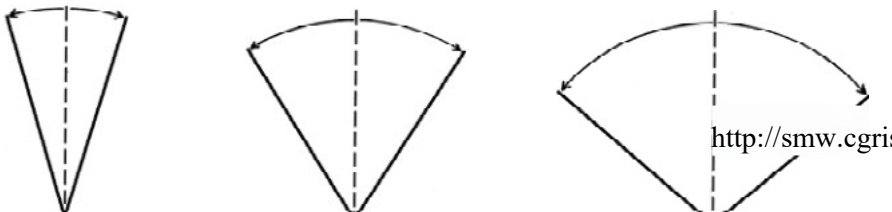
1 up-right

3 intermedius type

5 diffuse

7 hang down loosely

9 crawl



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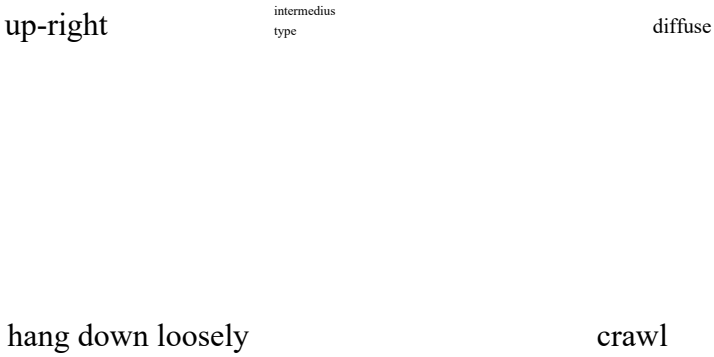


Figure 9: Stalk Angle

5.50 Stem Thickness

Stem diameter, expressed in mm.

5.51 The color of the stem nodes color
The color of the stem nodes.

1 Light green

2 purple

5.52 Stem internode color

The color of the stem internodes.

1 yellow

2 green

3 red

4 Purple line

5 purple

5.53 Stem and node exposure

The wrapping or exposed state of the stem nodes.

1 Bao

2 reveal

5.54 Branching capacity

The tillering capacity per rice plant.

1 Strong

5 China

9 weak

5.55 Ľıing

The degree of lodging of the stems.

1 straight

3 intermedius type

5 oblique

7 fall

9 Fu

5.56 Cap color

The color of the column head.

1 white

2 yellow

3 Light green

4 purple

5.57 Exposure rate of the column cap

When, after flowering, the stigma extends completely or partially beyond the glume, this is referred to as stigma exposure. Stigma exposure can be classified into two types: unilateral exposure and bilateral exposure. Unilateral exposure: one of the two stigmas of the glume flower extends beyond the glume; bilateral exposure: both stigmas of the glume flower extend beyond the glume.

The single-earlier exposure rate refers to the ratio of the number of individual ear-later florets that are exposed to the total number of florets that have opened; it is expressed as a percentage (%).

5.58 Double Exposed Rate of the Column Head

The ratio of the number of florets whose two staminal heads extend beyond the glume to the total number of florets that have opened, expressed as a percentage (%).

5.59 Total Exposed Rate of Column Heads

The ratio of the sum of the numbers of exposed single-and double-pedunculate glume flowers to the total number of opened glume flowers, expressed as a percentage.

5.60 Anther Shape

Shape of the anther.

1 Fine rod-shaped

2 water-stained appearance

3 Plump

5.61 Anther Color

Color of the anther.

1 White or milky white

2 popcorn

3 yellow

5.62 Anther length

Length of the anther. Unit: mm.

5. 63° opening angle

The angle between the inner and outer perianth when they are completely separated.

1 small

5 China

9 big

5.64 Opening Time

The time required for the perigynium to separate from the inner and outer perigynia and completely close; expressed in min.

5.65 Flower Blooming Period

The period from the opening of the first floret to the closure of the last floret during the peak flowering stage, expressed in hours.

5. 66 Flowering Height Peak

The period during which 20%–80% of the florets in the flowering population are in full bloom on that day.

1 morning

5 China

9 slow

5.67 Mangchang

The length of the awn growing from the outer tip.

1 not have

3 short

5 China

7 long

9 Extremely long

5.68 Mang Color

The color of the awns growing from the outer tips.

1 white

2 Stalk yellow

3 yellow

4 red

5 brown

6 purple

7 black

5.69 Mang distribution

The distribution of the bristles growing on the outer tips of the fronds.

1 not have

3 rare

5 few

7 China

9 More

5.70 Protective Hue

The color of the protective coating.

1 yellow

2 red

3 purple

5.71 **Protect the length of the hair**

The length of the hair guard.

1 short

3 China

5 long

7 Extremely long

9 asymmetric

5.72 Shape of the glume: mature grains exhibit a specific glume shape (Fig. 10).

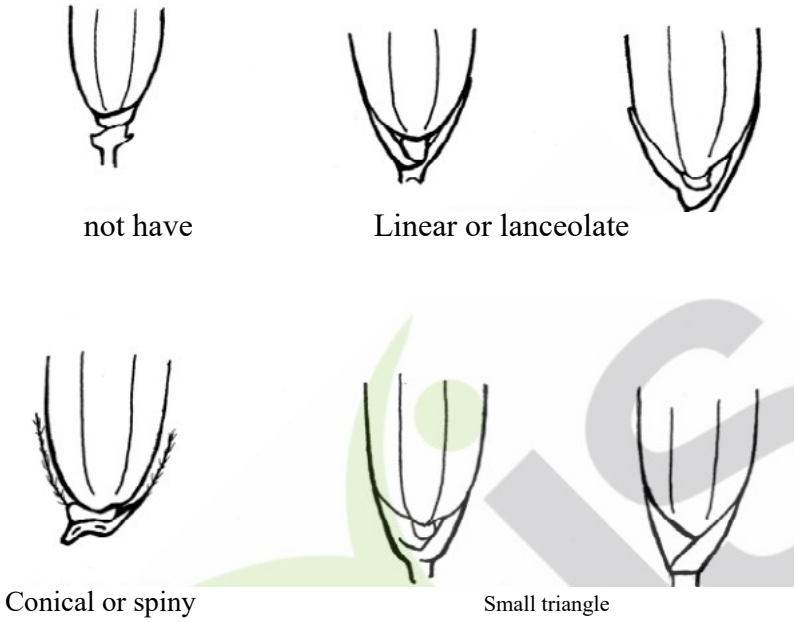


Figure 10: Shape of the Protective Cap

1 not have
2 Linear or lanceolate
3 Conical or spiny
4 Small triangle

5.73 Tip color

The color of the tip.

1 Stalk yellow
2 red
3 brown
4 purple
5 black

5.74 Ying Color

Color of the exaril and endaril of the grain.

1 yellow

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2 silver gray

3 brown

4 russet

5 atropurpureus

5.75 Glume Pubescence

Distribution of the hair growing on the surface of Ying.

1 not have

3 few

5 China

7 More

9 abundance

5.76 Particle Fallout

The ease or difficulty of grain shedding after the rice has reached full maturity.

1 Extremely low

3 low

5 China

7 Gao

9 polar altitude

5.77 Type of infertility

Refers to the types of infertility.

1 cytoplasm

2 nucleus

3 nucleo-cytoplasmic interaction

4 Light-temperature sensitive

5. 78% sterility rate

The percentage of sterile plants that fail to set fruit after bagging, expressed as a percentage of the total number of bagged plants.

5.79 Anthesis Dehiscence Degree

The degree of anther dehiscence during the flowering period.

1 No cracking

5 Few fissures

7 Central fissure

9 dehiscence

5.80 Pollinator infertility

The degree of pollen infertility.

1 Incomplete abortion

2 Severe infertility

3 Complete failure of fertilization

5.81 Pollen-induced abortion type

Classify according to the presence or absence of pollen and its morphology.

1 Non-pollen type

2 Classic failure type

3 Round failure type

4 mixed type

5 Partial senile type

6 Destructive type

5.82 The restorative capacity of the sterile line

The self-fertilization fruiting capacity of the F1 hybrids produced by crossing an infertile line with a typical restorer line; this is evaluated based on the fruiting rate.

1 weak

3 relatively weak

5 relatively strong

7 Strong

9 Extremely strong

5.83 The retention force of the retention system

The ability to maintain the sterility characteristics of the sterile line. This is evaluated by the self-pollination fruiting rate of the second-generation sterile line obtained through crossbreeding the maintainer line with the sterile line.

1 weak

3 relatively weak

5 stronger

7 Strong

9 Extremely strong

5.84 Restoration-type recovery capacity

The self-fertilization capability of the F1 hybrids obtained through test crosses between restorer lines and sterile lines is evaluated based on the self-fertilization rate.

1 weak

3 relatively weak

5 stronger

7 Strong

9 Extremely strong

5.85 Cross-pollination seed set rate in sterile lines

The seed set rate of male sterile lines when receiving exogenous pollen and undergoing cross-pollination.

1 low

3 lower

5 higher

7 Gao

9 polar altitude

5.86 affinity

The degree of compatibility with typical indica and japonica varieties is assessed by the fruit-set rate of the Fi plants after self-pollination following testcrossing with the test variety.

1 difference

5 China

7 good people

9 Excellent

5.87 Affinity profile

The types and scope of compatible germplasm resources within a germplasm resource possessing a certain degree of compatibility.

1 Ultra-narrow

5 Narrower

7 secondary

9 broad

6 Quality Characteristic

6.1 Brown rice rate

The percentage of unpolished rice after dehulling of clean paddy relative to the total grain weight, expressed as a percentage (%).

6.2 Refined rice rate

The percentage of milled polished rice relative to the total weight of the rice crop, expressed as a percentage (%).

6.3 Whole-Grain Refining Rate

The percentage of polished rice relative to the weight of the grain, expressed as a percentage (%).

6.4 Rice Grain Length

The maximum distance between the two ends of a polished rice grain. Unit: mm.

6.5 Fine Grain Width

The distance at the widest point of the polished rice grain. Unit: mm.

6.6 Long-to-Width Ratio of Polished Rice

The ratio of the length to the width of the polished rice grains.

6.7 Chalky White Grain Rate

The percentage of rice grains with chalky white color relative to the total number of grains in the sample, expressed as a percentage (%).

6.8 Chibi **small**

The percentage of the area of chalkiness in each rice grain relative to the total area of the grain, expressed as a percentage (%).

6.9 Chalkiness

chalkiness: the percentage of the total chalkiness area of the rice grains relative to the total grain area of the sample, expressed as a percentage (%).

6.10 pellucidity

The transparency of the polished rice grains, expressed as the relative light transmittance of the rice.

6.11 odour

The presence or absence of rice aroma.

1 Fragrant

2 Non-aromatic

6.12 gelatinization temperature

The temperature at which rice starch loses its crystallinity under heating conditions and undergoes irreversible expansion to form a starch paste is typically determined indirectly by the alkali erosion value method. The alkali erosion value refers to the extent to which an alkaline solution erodes refined rice grains and can indirectly indicate rice

The level of starch gelatinization temperature: the higher the alkali neutralization value, the lower the gelatinization temperature.

6.13 gel consistency

The flow length of rice gum after gelatinization of refined rice starch and subsequent cooling. Unit: mm.

6.14 Amylose Content

The percentage of直链淀粉 in polished rice relative to the dry weight of the polished rice, expressed as a percentage (%).

6.15 Crude Starch Content

The percentage of medium-to-high-starch content in polished rice relative to the dry weight of the polished rice, expressed as a percentage (%).

6.16 Crude protein content

The crude protein content in brown rice or polished rice expressed as a percentage of the dry weight of the brown rice or polished rice. Expressed in%.

6.17 Lysine content

The percentage of lysine content in brown rice or polished rice relative to the dry weight of the brown rice or polished rice, expressed as a percentage (%).

6.18 Crude fat content

The crude fat content in brown rice or polished rice expressed as a percentage of the dry weight of the brown rice or polished rice. Expressed in%.

7 Stress Resistance

7.1 Seedling-stage drought resistance

The ability of rice seedlings to tolerate or resist drought during the seedling stage. This is assessed using a comprehensive drought resistance coefficient.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.2 Drought resistance throughout the entire growth period

The ability of rice to tolerate or resist drought throughout its entire growth cycle. This is assessed using the drought resistance index.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.3 Salt Tolerance During the Germination Period

The ability of rice to tolerate or resist salt stress during the germination stage, assessed by the relative salt damage rate.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.4 Salt tolerance during the seedling stage

The ability of rice seedlings to tolerate or resist salt stress. This is assessed using the average leaf mortality rate or the salt damage index.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.5 Alkaline tolerance during the germination period

The ability of rice to tolerate or resist alkaline stress during the germination stage. This is assessed by measuring the germination rate after exposure to alkaline stress.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.6 Cold tolerance during the germination period

The ability of rice to tolerate or resist low temperatures during the germination period. This is assessed by the germination rate under low-temperature conditions.

1 Strong

5 China

9 weak

7.7 Cold tolerance during the bud stage

The ability of rice to tolerate or resist low temperatures during the seedling stage. This is assessed by the seedling mortality rate following low-temperature stress during the seedling stage.

1 Extremely Strong

3 Strong

5 China

7 weak

9 Extremely weak

7.8 Cold tolerance during the seedling stage

The ability of rice seedlings to tolerate or resist low temperatures during the seedling stage. This is assessed by the degree of red rot in the seedlings.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.9 Cold tolerance during the booting stage

The ability of rice to tolerate or resist low temperatures during the booting stage, evaluated by the empty-kernel rate.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.10 heat resistance

The ability of rice plants to tolerate or resist high temperatures. This is assessed by the empty-kernel rate following high-temperature stress during the flowering stage.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

8 Resistance to Pests and Diseases

8.1 Resistance to Bacterial Leaf Blight

During the rice seedling stage, the rice blast pathogen [*Magnaporthe grisea* (Hebert) Barr nov. comb. (anamorph

Pyricularia *oryzae* Cav.]'s resistance level.

1 highly resistance(HR)

2 resist(R)

3 moderate resistance(MR)

4 Zhonggan 1 (MS1)

5 Zhonggan 2 (MS2)

6 Sensation 1 (S1)

7 Sensitivity 2 (S2)

8 High Sensitivity 1 (HS1)

9 High Sensitivity 2 (HS2)

8.2 Resistance to Leaf Blight

The resistance of rice to the rice blast fungus during the tillering stage *[Magnaporthe grisea (* Hebert) Barr nov. comb. [anamorph Pyricularia *oryzae Cav.]].*

1 highly resistance(HR)

2 resist(R)

3 moderate resistance(MR)

4 Zhonggan 1 (MS1)

5 Zhonggan 2 (MS2)

6 Sensation 1 (S1)

7 Sensitivity 2 (S2)

8 High Sensitivity 1 (HS1)

9 High Sensitivity 2 (HS2)

8.3 Resistance to穗颈瘟

From the heading stage to maturity, the resistance of the panicle neck to the rice blast fungus [*Magnaporthe grisea (* Hebert) Barr nov. comb. (anamorph Pyricularia *oryzae Cav.)* varies in strength.

1 highly resistance(HR)

2 resist(R)

3 moderate resistance(MR)

5 moderate susceptibility(MS)

7 feel(S)

9 High Sensitivity (HS)

8.4 Resistance to穗节瘟

The resistance of the spikelet nodes of rice from the heading stage to maturity to the rice blast pathogen *[Magnaporthe grisea (* Hebert) Barr nov. comb. (anamorph Pyricularia *oryzae Cav.)* varies in strength.

1 highly resistance(HR)

2 resist(R)

3 moderate resistance(MR)

5 moderate susceptibility(MS)

7 feel(S)

9 High Sensitivity (HS)

8.5 Resistance to white leaf blight

The resistance of rice to *Xanthomonas oryzae (* Uyeda et al.) Dowson.

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 Zhong Kang (MR)

7 feel(S)

9 High Sensitivity (HS)

8.6 Resistance to wood rot

The resistance of rice to the rice stripe blight pathogen (*Thanatephorus sasakii (* Shirai) Tu and Kimbrough).

0 immunity(IM)

1 resist(R)

3 moderate resistance(MR)

5 moderate susceptibility(MS)

7 feel(S)

9 High Sensitivity (HS)

8.7 Resistance to Bacterial Stripe Disease

The resistance of rice to Xanthomonas *oryzicola Fang et al.* varies in strength.

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 moderate susceptibility(MS)

7 feel(S)

9 High Sensitivity (HS)

8.8 Resistance to the white-backed leafhopper

The resistance of rice to the white-backed leafhopper [*Sogatella furcifera (* Horvath)], a rice pest,

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 moderate susceptibility(MS)

9 feel(S)

8.9 Brown planthopper resistance

The resistance of rice to the rice-specific pest, the brown planthopper [*Nilaparvatalugens* (Stdl)].

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 moderate susceptibility(MS)

9 feel(S)

8.10 Resistance to rice galls

The resistance of rice to the rice stem borers *(Orseolia oryzae Wood et Mason)*.

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 moderate susceptibility(MS)

9 feel(S)

8.11 Resistance to the second-generation caterpillar

The resistance of rice to the two-spined stem borer [*Chilo supperssalis (* Walker)]

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 moderate susceptibility(MS)

9 feel(S)

8.12 Resistance to the third-generation caterpillar

The resistance of rice to the three-leaf stem borer [*Tryporyza incertulas (* Walker)]

0 immunity(IM)

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 moderate susceptibility(MS)

9 feel(S)

9 Other Characteristics and Features

9.1 Use

Specific uses of rice.

1 Edible rice

2 Health Rice

3 Food-processing rice

4 Industrial raw rice

5 Feed-grade rice

9.2 Fingerprinting and Molecular Markers

9.3 Remarks

A detailed description of the specific descriptors or codes for rice germplasm.